Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2024

Course: B.Tech
Code: 100403

Subject: Digital Electronics

Time: 03 Hours
Full Marks: 70

Instructions: -

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Answer of the following questions (any seven only):- [2 x 7 = 14]

(a) What is the use of K-map?
(b) Discuss Universal gates.
(c) Add hexadecimal number 2ABC & 98F2.
(d) Number of 2:1 mux requires designing 256:1 mux is ....
(e) Find 2's complement of 1011011.
(f) What is the function of a sample-and-hold circuit?
(g) Define term propagation delay.
(h) Define race around condition in JK flip flop.
(i) What is the purpose of expanding memory size?
(j) Differentiate between ROM and RAM.

Q.2 (a) Realize XNOR logic function using NAND gate only. [7]
(b) Simplify $Y = ABC + AB\overline{C} + A\overline{B}C$ [7]

Q.3 A logic circuit has four inputs A,B,C,D and output Y. Y=1 when A & B are both 1,subjected to the condition that C and D are both low or both high. Design the logic circuit. [14]

Q.4 (a) Design 8 to 3 line Encoder circuit. [7]
(b) Implement NAND gate using TTL logic family. [7]

Q.5 (a) Design a parallel-to-serial converter using shift registers and explain its operation. [7]
(b) Compare the characteristics and applications of JK and T flip-flops. [7]

Q.6 (a) Explain the working principle of an R-2R ladder DAC with a detailed diagram. [7]
(b) Design a 3-bit flash ADC and explain its working with an example. [7]

Q.7 (a) Find the Simplified logical expression for Y. [7]
$Y(A, B, C, D, E) = \sum m(0,2,4,7,8,10,12,16,18,20,23,24,25,26,27,28)$
(b) Summarize the design procedure for a synchronous sequential circuit. [7]

Q.8 (a) Implement S-R, T, D flip-flops using J-K flip-flop. Also show the implementation with help of State Tables. [7]
(b) Discuss the organization and operation of content-addressable memory (CAM). [7]

Q.9 Write short notes on the following: [7x2=14]
(a) Binary Parallel Adder
(b) Digital IC logic families

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